

TEST PLAN

1. INTRODUCTION:

This test plan explains how the given API will be tested. The API is designed to return the second largest number from a list (array) of numbers provided in the request. The purpose of this testing is to make sure the API works correctly for different types of inputs and handles invalid cases properly.

2. OBJECTIVE:

The objective of this testing is to verify that the API correctly returns the second largest number from the array. Along with that, the API will also be checked to see how it behaves when incorrect or invalid data is provided.

3. SCOPE OF TESTING:

In this testing task, the main focus will be on validating the functionality of the API. The following scenarios will be tested:

- Checking if the API returns the correct second largest number.
- Testing with different input values such as positive numbers, negative numbers, and duplicate numbers.
- Testing invalid inputs like empty arrays or incorrect data types.
- Verifying authorization using the provided bearer token.

The testing will be performed using **Postman** to send requests and check responses.

4. TESTING APPROACH:

During testing, different approaches will be used to verify the API functionality.

- **FUNCTIONAL TESTING:**
To check whether the API returns the correct result when valid input is provided.
- **NEGATIVE TESTING:**
To verify how the API behaves when invalid or unexpected inputs are given.
- **AUTHORIZATION TESTING:**
To check whether the API allows access only when a valid token is provided.

5. TEST ENVIRONMENT:

COMPONENT	DETAILS
Testing Tool	Postman
API Method	POST
System	Windows
Internet	Required

6. TEST DATA:

Different input values will be used while testing the API, for example:

- [1,2,3,4,5]
- [10,20,30]
- [-1,-2,-3,-4]
- [5,5,5]
- []

These inputs will help test both valid and invalid scenarios.

7. TEST DELIVERABLES:

The following documents will be submitted as part of this assignment:

- Test Plan document.
- Test Cases sheet.
- Screenshots of API testing.

8. CONCLUSION:

This testing will help verify whether the API is working as expected. By testing different scenarios, we can ensure that the API returns the correct result and handles incorrect inputs properly.